

Where Does the Relative Clause Attach? Probing Prosodic and Semantic Sensitivity in Large Audio Language Models.

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Abstract

Despite extensive research on human sentence processing, little is known about whether large audio language models (LALMs) exhibit similar sensitivity to prosodic and semantic cues during syntactic ambiguity resolution. We evaluated the ability of five state-of-the-art LALMs to resolve relative clause (RC) attachment ambiguities across three typologically diverse languages: English, Chinese, and Arabic. We constructed audio stimuli with two types of attachment cues: (1) prosodic cues, implemented through manipulated boundary placement and prosodic prominence to favor either high or low attachment, and (2) semantic cues, in which world knowledge favored one attachment site. Each stimulus was probed using both yes/no and forced-choice prompts. Our findings revealed limited and inconsistent sensitivity to prosodic cues, with low attachment remaining the dominant interpretation. In contrast, models demonstrated greater sensitivity to semantic and world knowledge cues. Forced-choice prompts yielded more stable attachment preferences and fewer uncertain responses than yes/no prompts. These results raise questions about the cognitive capacities of current LALMs, suggesting that their syntactic processing is driven primarily by semantic information, with prosodic cues playing a limited and inconsistent role in ambiguity resolution.

1 Introduction

Research in psycholinguistics seeks to explain how comprehenders construct syntactic representations and resolve ambiguity during language processing, integrating structural, semantic, and pragmatic information (Rah, 2010; Havik et al., 2009). A particularly well-known case of syntactic ambiguity is **relative clause (RC) attachment ambiguity**, in which a relative clause follows a complex noun phrase with two possible antecedents (Fedorova et al., 2007; Lee & De Santo, 2024). For example, in *The doctor met the son of the man who had a beard*, the relative clause (*who had a beard*) may modify either the more distant NP1 (*the son*), yielding a **high-attachment interpretation**, or the adjacent NP2 (*the man*), yielding a **low-attachment interpretation** (see Figure 1).

Although both attachment options are grammatically available, speakers exhibit systematic preferences in resolving RC attachment ambiguity. Languages differ in their default attachment preferences. English speakers, for example, exhibit a default preference for low attachment, whereas Spanish speakers tend to prefer high attachment (Cuetos & Mitchell, 1988; Gilboy et al., 1995; De Pedis et al., 2025). Such cross-linguistic variation in RC interpretation has been attributed to the interplay between universal parsing mechanisms and language-specific processing strategies. One influential account is the Late Closure principle, proposed within the Garden Path Theory (Frazier & Fodor, 1978), which proposes that incoming constituents are preferentially attached to the phrase currently being processed. Applied to RC attachment, the principle predicts a preference for low attachment, consistent with the view that local attachment minimizes processing effort and places fewer demands

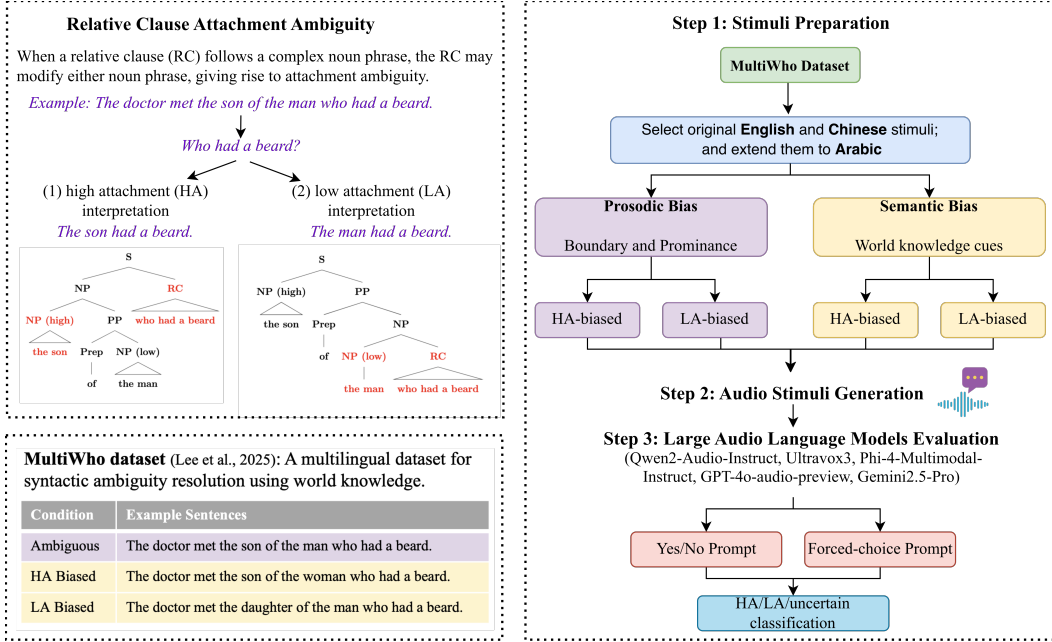


Figure 1: Overview of the study design for testing LALMs’ prosodic and semantic sensitivity in RC attachment ambiguity resolution. The left panel illustrates RC attachment ambiguity and the MultiWho dataset, and the right panel shows the experimental workflow.

on working memory (Papadopoulou, 2006). In contrast, parametrized models such as Predicate Proximity propose a more flexible mechanism, in which attachment is influenced by the relation between the main predicate and the candidate antecedents, accounting for the high-attachment preference observed in some languages (Papadopoulou & Clahsen, 2003). Beyond these parsing accounts, Fodor’s Implicit Prosody Hypothesis (Fodor, 1998; 2002) argues that readers generate an internal prosodic contour even during silent reading and that this implicit prosody can bias attachment decisions. Empirical studies further demonstrate that RC attachment preferences are shaped by multiple interacting linguistic cues. Lexical plausibility, animacy, constituent length, prosodic phrasing, and information structure have all been shown to influence how readers and listeners resolve RC attachment ambiguity (Desmet et al., 2006; Kwon et al., 2019; Clifton Jr et al., 2002; Jun, 2003; Hemforth et al., 2015).

Accordingly, scholarly attention has increasingly turned to the question of whether large language models (LLMs) possess the capacity to resolve RC attachment ambiguity (e.g., Lee et al., 2024; Kamerath & De Santo, 2025a; Issa & Atouf, 2024; Kamerath & De Santo, 2025b). Such inquiry is theoretically motivated, as it affords the opportunity to evaluate the cognitive capacities of LLMs against the same empirical standards employed in the assessment of human language processing. Beyond these text-based models, large audio language models (LALMs) have emerged as a class of systems that integrate audio representations with LLMs to enable unified interaction over spoken content, environmental sounds, and even music (Chu et al., 2023; 2024; Ding et al., 2025; Tang et al., 2023). The concept of understanding in LALMs has broadened to encompass not only speech but also non-speech audio, music, and other acoustic signals. These capabilities have been the subject of systematic investigation across an expanding range of benchmarks, including AudioBench (Wang et al., 2025a), VoiceBench (Chen et al., 2026), SD-Eval (Ao et al., 2024), ADU-Bench (Gao et al., 2025), and MMAU (Wang et al., 2025b). Each benchmark targeted distinct dimensions of audio comprehension, such as instruction following, spoken dialogue understanding, or multi-task audio reasoning. Despite this recent interest in understanding the capabilities of LALMs, their application to human language processing remains largely unexplored, and to the best of our knowledge, no study has yet examined the capacity of LALMs to resolve RC attachment ambiguity.

The goal of this study is to investigate whether LALMs demonstrate human-like language processing abilities in resolving RC attachment ambiguities across three languages (English, Chinese, and Arabic). In particular, we examine the extent to which prosodic cues (e.g., boundary placement and prominence) and semantic cues (e.g., contextual and world knowledge biases) guide models’ attachment preferences between the two potential noun phrase antecedents. Our key contributions are: (1) we extended the existing MultiWho dataset to a new language, Arabic, and made this resource publicly available to support future research, (2) we systematically evaluated LALMs on their sensitivity to prosodic and semantic cues in resolving relative clause attachment ambiguity, and (3) we conducted a cross-linguistic investigation across English, Chinese, and Arabic; we show that LALM behavior is asymmetric: prosodic cues have weak and inconsistent effects, while semantic cues influence responses more strongly but still interact with a persistent low-attachment bias.

2 Related work

2.1 LALMs Benchmarks

There has been considerable work that systematically benchmark LALMS. For example, AudioBench (Wang et al., 2025a) is a comprehensive benchmark that encompasses 8 distinct task categories and 26 datasets in addition to investigating prompt sensitivity in LALMs. AIR-Bench (Yang et al., 2024) evaluates instruction-following capabilities of LALMs across multiple audio modalities including speech, environmental sounds, and music. AIR-Bench is structured along two dimensions: a foundation benchmark consisting of 19 tasks with approximately 19,000 single-choice questions, and a chat benchmark with approximately 2,000 open-ended question-answer instances. The latter is evaluated using GPT-4 as the judge, comparing generated responses to reference answers. The MMAU (Sakshi et al., 2025) focuses on testing audio understanding and reasoning across three domains: speech, sound, and music. MMAU frames all tasks as multiple-choice questions, enabling fully automated evaluation without human annotation or language model judges. Dynamic-SUPERB (Huang et al., 2024a;b) is a two-phase benchmark designed for evaluating universal instruction-based speech models in a zero-shot fashion. The evaluation results confirm that no model performs well universally, showing that SALMONN-13B excels in English ASR while Qwen2-Audio-7B-Instruct shows strong emotion recognition, but neither generalizes across the full task range.

Some benchmarks have investigated content understanding. For example, SD-Eval (Ao et al., 2024) focuses on a narrower but critical dimension of LALM capability: spoken dialogue understanding beyond the literal words. The results indicated that LALMs fall short of generating contextually appropriate responses, an issue attributed to the absence of principled task definitions and suitable evaluation datasets. Similarly, ADU-Bench (Gao et al., 2025) benchmarks LALMs on open-ended audio dialogues spanning 3 general scenarios, 12 skills, 9 multilingual languages, and 4 categories of ambiguity handling beyond the literal meaning. Experiments across 16 LALMs revealed that models perform better on information-seeking questions than on everyday conversational exchanges, and consistently fail at handling roleplay scenarios, non-English languages, and phonetic ambiguities.

Other benchmarks have specifically examined the problem of audio hallucination. Ghosh et al. (2023) introduced Comp-A to evaluate compositional reasoning, finding that current models perform only marginally better than random chance when tasked with matching audio clips to captions that differ in the order or attribute-binding of acoustic events. Sung-Bin et al. (2024) introduced the AVH-Bench, a comprehensive cross-modal hallucination benchmark for audio-visual LLMs. Results revealed that most existing models struggle to discern subtle relationships between audio and visual signals. In addition, AHa-Bench (Cheng et al., 2025a) categorized audio hallucinations into three distinct types: semantic, acoustic, and semantic-acoustic; and evaluated seven open-source LALMs. The results exposed critical limitations in models’ ability to jointly interpret semantic and acoustic information.

2.2 Relative clause attachment ambiguity resolution

RC attachment ambiguity arises when a relative clause can modify either of two candidate noun phrases inside a complex noun phrase. Previous research has shown cross-linguistic variation in attachment preferences (Cuetos & Mitchell, 1988; Grillo & Costa, 2014). Native speakers of English have been reported to prefer low attachment — linking the relative clause to the structurally local, lower NP — whereas speakers of Spanish show a robust default preference for high attachment, favoring modification of the structurally higher, non-local head noun (Gilboy et al., 1995). This high-attachment preference has been observed across a range of languages, including French (Zagar et al., 1997), Korean (Jun, 2004), and Japanese (Kamide & Mitchell, 1997). Chinese has been claimed to favor low attachment despite its head-final relative clause structure (Shen, 2006; Kwon et al., 2019), though here the relevant notion is structural rather than linear locality, since the relative clause precedes both candidate NPs. Arabic occupies a contested position. Abdelghany & Fodor (1999) reported a low-attachment preference, consistent with the Implicit Prosody Hypothesis. Bidaoui et al. (2016), in contrast, reported high attachment among native speakers and attribute this to Predicate Proximity: because Arabic permits flexible word order where verbs can be distant from their complements, the parser may be more “activated” to make long-distance attachments to the structurally higher head noun. This unresolved status is one of our motivations for including Arabic in the present study. LALMs’ behavior on Arabic provides an informative third data point against which to assess whether models default to a low-attachment bias universally or track the variability observed in the human literature.

However, these preferences could be modulated by multiple factors such as lexical constraints such as animacy and concreteness of noun phrase (Desmet et al., 2006; Kwon et al., 2019), and syntactic properties like position and length of the RC (Hemforth et al., 2015). Prosody and semantics provide two additional sources of information. The Implicit Prosody Hypothesis (Fodor, 1998; 2002) proposes that comprehenders project a default prosodic contour during silent reading and that this internally generated prosodic structure influences syntactic parsing decisions. In the case of RC attachment, this matters because prosodic phrasing can make one attachment analysis sound more natural than the other. A boundary immediately before the relative clause, that is after NP2 in English examples, tends to support high attachment; a boundary after NP1 tends to support low attachment (Clifton Jr et al., 2002; Fodor, 2002; Hwang et al., 2011; Jun, 2003). For example, in the sentence *Someone shot the servant of the actress # who was on the balcony*, the prosodic boundary # before the RC biases interpretation toward high attachment (i.e., the servant was on the balcony). Prosodic cues like prominence also plays an important role. For instance, Schafer (1996) showed that accented noun phrases are more likely to be interpreted as the head of the RC. Additionally, semantic and world knowledge guide attachment: comprehenders prefer the more plausible interpretation based on their world knowledge, often overriding structural preferences (MacDonald et al., 1994). For example, in the sentence *The doctor met the son of the woman who had a beard*, semantic plausibility favors interpreting the relative clause as modifying *the son*, because having a beard is more plausible for the son than for the woman. This yields a high-attachment interpretation, even though a structurally local attachment preference would favor NP2, *the woman*.

Recent LLM work uses RC attachment to test whether prompted model judgments resemble human parsing preferences. Lee et al. (2024) reported that text-based LLMs show a strong low-attachment preference across languages and only partial sensitivity to cross-linguistic variation. Their later MultiWho study (Lee et al., 2025) showed that LLMs can respond to world-knowledge bias, but often do so rigidly and without the flexibility seen in human interpretation. Scheinberg et al. (2025) focused on English and showed that semantic biases and world-knowledge plausibility can shift some models away from low attachment, but that this override is inconsistent and depends on model family and prompt format. What remains unclear is whether LALMs behave differently when the disambiguating information is present in the audio signal. If LALMs use prosodic representations in a way analogous to the mechanisms proposed by the Implicit Prosody Hypothesis, their RC attachment preferences should shift systematically in response to prosodic boundary and prominence manipulations in the input audio. If LALMs integrate semantic and world-knowledge cues

into syntactic ambiguity resolution, their responses should also shift toward the attachment interpretation made more plausible by the semantic bias.

3 Methods

3.1 Dataset

The experimental stimuli were drawn from the MultiWho dataset (Lee et al., 2025), a multilingual corpus across six typologically diverse languages designed to evaluate how humans and LLMs resolve RC attachment ambiguity. For each language, the dataset contains 96 sets, each including three sentences corresponding to the ambiguous, high-attachment-biased (HA-biased), and low-attachment-biased conditions (LA-biased). Table 1 shows an English example set for the three conditions. In the ambiguous condition, both NP1 and NP2 are plausible antecedents of the relative clause. In the HA- and LA-biased conditions, semantic plausibility biases attachment toward NP1 and NP2, respectively. In our experiments, the ambiguous sentences provide the baseline items and the source text for prosodic manipulations, whereas the original HA- and LA-biased sentences are used to test semantic and world-knowledge bias.

Conditions	Sentences
Ambiguous	The doctor met the son _{NP1} of the man _{NP2} <i>who had a beard</i> _{RC}
HA-Biased	The doctor met <i>the son</i> _{NP1} of the woman _{NP2} <i>who had a beard</i> _{RC}
LA-Biased	The doctor met the daughter _{NP1} of <i>the man</i> _{NP2} <i>who had a beard</i> _{RC}

Table 1: Example English stimuli for relative clause attachment conditions.

In the present study, we selected the English and Chinese subsets of the MultiWho dataset. To broaden the cross-linguistic comparison, we additionally included Arabic, a less-studied language in research on syntactic ambiguity and a relatively low-resource language in current LLM training data. The original English sentences were translated into Arabic by the second author, a native speaker of Arabic, to ensure grammaticality and naturalness. English, Chinese, and Arabic differ in the structural realization of relative clauses, including the position of the relative clause with respect to the head noun (e.g., prenominal in Chinese vs. postnominal in English and Arabic) and canonical word order. This allows us to examine whether LALMs are sensitive to cross-linguistic structural differences beyond surface-level word order.

3.2 Stimuli design

To test LALM’s sensitivity to prosodic and semantic cues, we used the MultiWho conditions in two experiments. In the prosody experiment, ambiguous sentences served as the neutral baseline and were also synthesized with HA- and LA-biased prosody. In the semantic experiment, the original HA- and LA-biased sentences were used to test whether world knowledge shifts attachment interpretation. All stimuli were converted into audio as input for LALM evaluation.

Prosodically biased sentences Following prior work showing that prosodic boundary placement and prominence influence attachment preferences (Clifton Jr et al., 2002; Fodor, 2002), we constructed prosodically biased variants from the ambiguous sentences (see 1a below) in the MultiWho dataset. The original ambiguous sentences served as the neutral baseline (96 sentences per language), and each was additionally synthesized in two prosodically biased conditions: HA-biased and LA-biased. In the HA-biased condition (see 1b below), a prosodic boundary was placed before the relative clause (after NP2), with prominence assigned to NP1, biasing interpretation toward high attachment. In the LA-biased condition (see 1c below), the boundary was placed after NP1, with prominence assigned to NP2, biasing interpretation toward low attachment. Importantly, the lexical content was identical across conditions; only the intended prosodic realization differed. Here, # denotes a prosodic boundary, and **boldface** denotes prominence:

- (1a) The doctor met the son of the man who had a beard. (ambiguous)
 (1b) The doctor met the **SON** of the man # who had a beard. (prosodically HA-biased)
 (1c) The doctor met the son # of the **MAN** who had a beard. (prosodically LA-biased)

Semantically biased sentences For semantic cues, we used the original HA- and LA-biased sentences from the MultiWho dataset (192 sentences per language). Unlike the prosodic variants, these stimuli were not created by changing the audio realization of an otherwise identical sentence. Instead, the lexical content itself creates the bias: world knowledge makes either NP1 or NP2 the more plausible antecedent. The cues draw on different types of world knowledge, including gender-based plausibility (e.g., *giving birth* typically associates with females), age-based plausibility (e.g., *the brother of the baby who was driving a car*), and profession-based plausibility (e.g., *the interpreter of the judge who was monolingual*). By making one candidate antecedent more plausible than the other, these cues bias the intended attachment interpretation toward either NP1 or NP2.

Audio stimuli generation Following established practice in recent LALM benchmarking and evaluation studies that use synthesized audio stimuli (Cheng et al., 2025a;b; Fang et al., 2025), we generated all audio stimuli using the ElevenLabs text-to-speech (TTS) system with different prosodic realizations. The synthesis procedure differed for the two stimulus types: (1) For the prosody experiment, we followed ElevenLabs’ prompting guidance to elicit the intended boundary placement and prominence. Ambiguous sentences were synthesized in three versions: a neutral baseline and two prosodically biased versions, with the latter designed to realize HA- and LA-biased prosodic cues. (2) For the semantic experiment, the original HA- and LA-biased MultiWho sentences were synthesized with natural prosody, without additional boundary or prominence manipulation. The resulting audio files were used as input to evaluate the LALMs.

After synthesis, we checked the generated audio stimuli both qualitatively and acoustically. Native speakers of each language judged stimuli to be intelligible and generally natural. We also conducted an acoustic verification analysis which confirms that the intended manipulations were acoustically realized. Specifically, we analyzed 15 randomly sampled triplets per language (neutral baseline, prosodically HA-biased, and prosodically LA-biased; 135 files total) using Parselmouth (Jadoul et al., 2018). We measured pause duration at the intended boundary locations, peak F0 on the head nouns of NP1 and NP2 (normalized to semitones relative to the utterance-level median F0), and mean intensity for each target NP. We compared the HA and LA conditions within matched triplets using paired Wilcoxon signed-rank tests. Prominence cues were reliably realized in all three languages (English, Arabic, and Chinese), while boundary cues, measured by pause duration, were reliably realized in English and Arabic. In Chinese, no detectable silent boundaries were observed; instead, the manipulation was realized primarily through prominence. Full methodology and statistical results are reported in Appendix A.

3.3 Experimental procedure

Language Models We tested five models (both open-source and proprietary systems), including Qwen2-Audio-Instruct¹ (Chu et al., 2023), Ultravox², Phi-4-Multimodal-Instruct³ (Abouelenin et al., 2025), GPT-4o-audio-preview (OpenAI, 2024), and Gemini2.5-Pro (Google DeepMind, 2025), to evaluate how LALMs resolve RC attachment ambiguities. Appendix B presents details about the open-source models.

Prompt Design For each audio stimulus, we used two prompt formats to elicit model responses: a yes/no question and a forced-choice question (see Figure 2). The yes/no prompt asked the model to verify whether a given referent satisfied the property expressed by the relative clause. The forced-choice prompt asked the model to choose explicitly between NP1 and NP2 as the antecedent of the relative clause. Thus, the two prompt

¹<https://huggingface.co/Qwen/Qwen2-Audio-7B-Instruct>

²<https://huggingface.co/fixie-ai/ultravox-v0.5-llama-3.1-8b>

³<https://huggingface.co/microsoft/Phi-4-multimodal-instruct>

formats test the same underlying attachment interpretation but differ in the response form they require: binary verification versus explicit attachment selection.

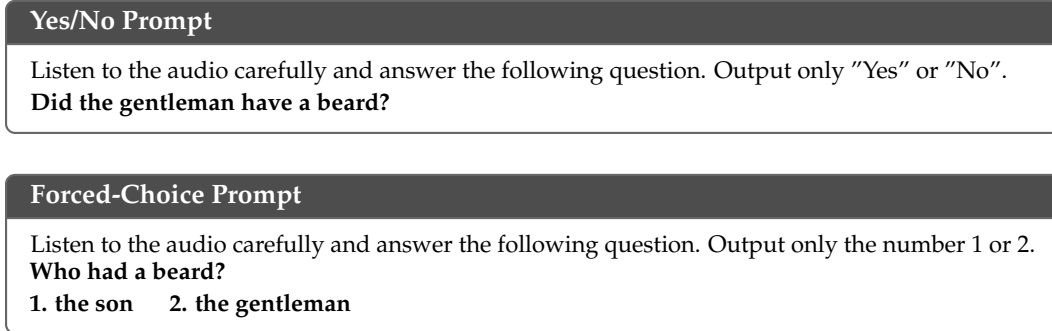


Figure 2: The yes/no and forced-choice prompts for the same audio stimulus *The doctor met the son of the gentleman who had a beard.*

Data analysis We mapped each model response to one of three outcome categories: *high attachment (HA)*, *low attachment (LA)*, or *uncertain*. A response was coded as HA if the model selected the non-local NP1 as the antecedent, and as LA if it selected the local NP2. If the model did not clearly identify either antecedent, the response was coded as *uncertain*. For each condition, language, model, and prompt format, we computed the proportion of HA, LA, and uncertain responses. This allowed us to assess whether models followed the intended bias of each condition, producing HA responses for HA-biased items and LA responses for LA-biased items.

4 Results

4.1 LALM’s sensitivity to prosodic manipulation/information in ambiguous cases

Yes-no prompt results As shown in Figure 3, attachment preferences varies substantially across models and languages in the baseline (ambiguous) condition. In English, several models (e.g., Gemini and GPT) favored low attachment, whereas others (e.g., phi_instruct and ultravoX) show a strong preference for high attachment. In contrast, models generally showed stronger high-attachment preferences in Chinese and Arabic. Moreover, under prosodic manipulation, models showed limited and inconsistent sensitivity to prosodic cues. Although prosodic boundary and prominence are designed to bias high- or low-attachment interpretations, observed shifts are partial and vary across models. While some models (e.g., Gemini and Phi-Instruct) demonstrate greater sensitivity to the prosodic manipulation, others (e.g., qwen2-instruct) frequently produce uncertain responses, indicating limited use of prosodic information when resolving attachment ambiguity.

Forced choice prompt results Compared with the yes/no prompt, the forced-choice prompt virtually eliminates uncertain responses and produces more consistent attachment decisions across models. In the baseline (ambiguous) condition, most models exhibit a stronger preference for low attachment than in the yes/no condition, although some language- and model-specific variation remains (e.g., UltraVox in Arabic). Under prosodic manipulation, changes in attachment preference are generally limited in the high-attachment condition. Although some models (e.g., Phi-Instruct) show increased high-attachment responses, low-attachment preference remains the dominant response overall. In contrast, the low-attachment condition leads to strong and consistent shifts toward low attachment across models. As a result, prosodic effects are more robust for low-attachment than for high-attachment interpretations. Overall, forcing a decision reduces uncertainty and leads to more consistent attachment responses, whereas the influence of prosodic manipulation remains relatively modest.

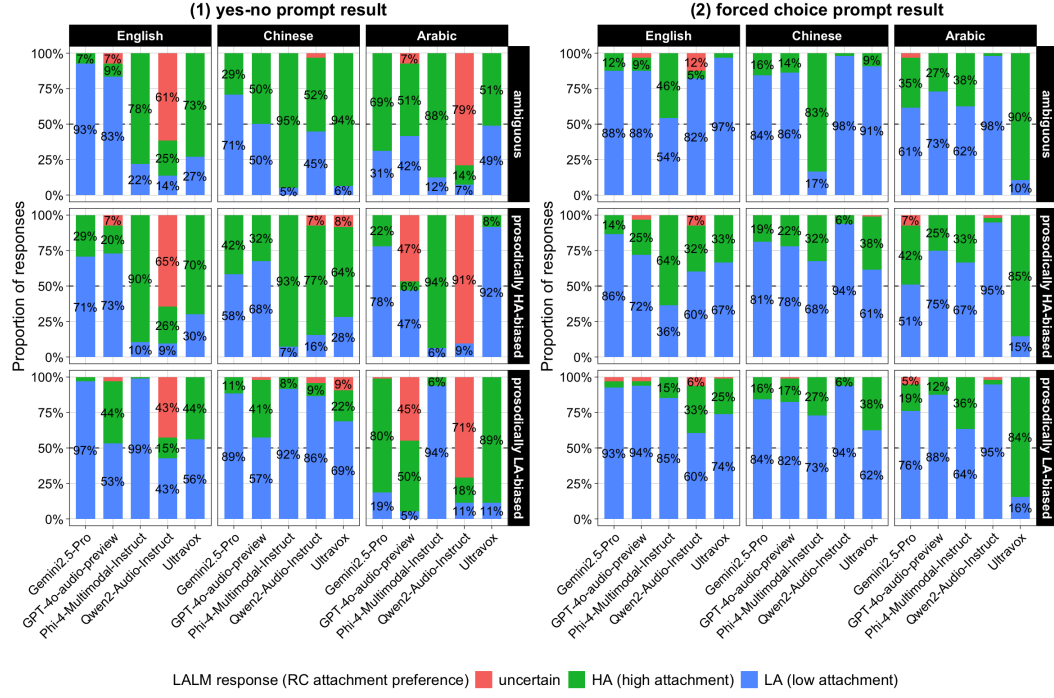


Figure 3: Distribution of LALM responses under prosodic manipulation. Rows represent the three prosody conditions, columns represent the three languages, and stacked bars show the proportions of HA, LA, and uncertain responses for the yes-no (left) and forced-choice (right) prompts.

4.2 LALM’s sensitivity to world knowledge and semantics bias in unambiguous cases

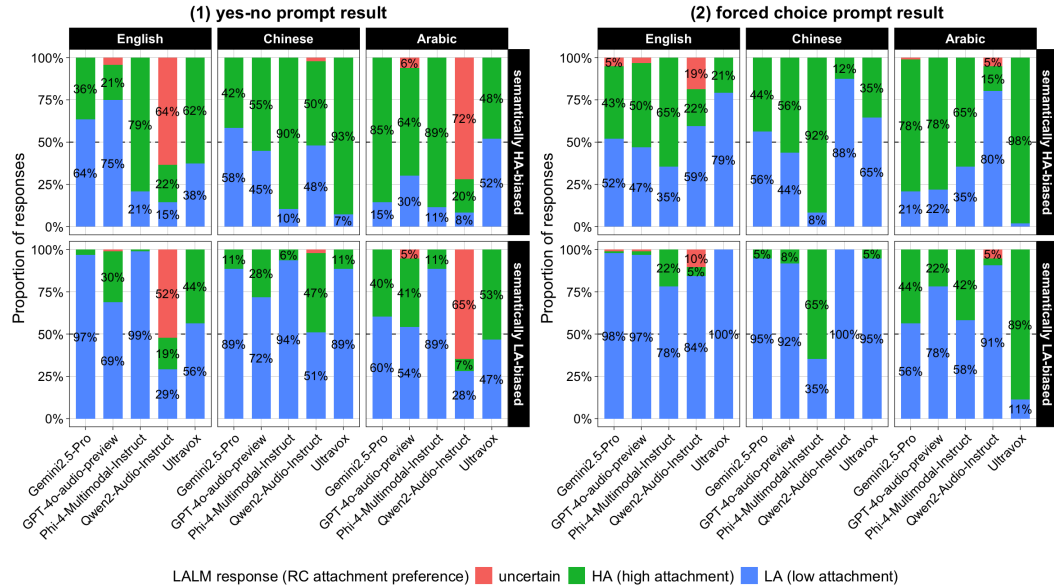


Figure 4: Distribution of LALM responses under semantic (world knowledge) bias

Yes-no prompt results Figure 4 shows that world knowledge or semantic bias affects model responses more strongly than prosodic bias, but again the effect is asymmetric. In the HA-biased semantic condition, models more often produce HA interpretations in Chinese and Arabic than in English. Notably, some models (e.g., Gemini and GPT) maintain a strong low-attachment preference in English even when processing high-attachment-biased sentences. In contrast, in the LA-biased semantic condition, models much more consistently produce low-attachment interpretations. Qwen2-Audio-Instruct again produces many uncertain responses in the yes/no prompt format, especially in English and Arabic. These patterns suggest that semantic plausibility can influence LALM attachment judgments, but high-attachment semantic cues do not consistently override the low-attachment default.

Forced choice prompt results When models are forced to choose between the two candidate antecedents, uncertain responses are minimal and the semantic asymmetry becomes clearer. HA-biased semantic items elicit strong high-attachment responses in Arabic and weaker alignment in English. In Chinese, model behavior is mixed, with Phi-Instruct and GPT favoring high attachment, whereas other models remain closer to low attachment. By contrast, LA-biased semantic items elicit highly consistent low-attachment responses across most models and languages, with only a small number of model-language exceptions (i.e., phi-instruct in Chinese and ultravox in Arabic). The contrast between the two semantic directions indicates that world knowledge is available to the models, but its influence is strongest when it supports low attachment.

5 Discussion

Across our experiments, LALMs exhibit asymmetric sensitivity to linguistic cues during RC attachment ambiguity resolution. In English, Chinese, and Arabic, models showed a strong preference for low attachment, even when prosodic or semantic cues bias interpretation toward high attachment. This pattern is consistent with prior work showing a robust low-attachment bias in LLMs across typologically diverse languages (Lee et al., 2024; 2025). While semantic cues influence attachment preferences more reliably than prosodic cues, neither cue type consistently overrides this default low-attachment preference. Forced-choice prompting substantially reduces uncertainty and produces more decisive responses, but it does not eliminate the underlying low-attachment bias. These findings suggest that current LALMs rely on linguistic cues in a constrained and asymmetric manner rather than flexibly integrating multiple sources of information in a human-like manner.

Despite explicit manipulation of prosodic boundaries, models showed inconsistent sensitivity to prosodic cues, with cues favoring low attachment proving more effective than those favoring high attachment. These findings provide only limited support for the Implicit Prosody Hypothesis (Fodor, 1998; 2002) in the context of LALMs. Whereas human listeners reliably exploit prosodic structure to guide syntactic disambiguation (Clifton Jr et al., 2002; Hwang et al., 2011; Jun, 2003), the present results suggest that LALMs do not robustly encode or utilize prosodic representations during sentence processing. Several non-mutually exclusive factors across the LALM processing pipeline may contribute to this prosodic insensitivity. First, current LALMs are trained predominantly on ASR-style audio-text pairs, where the training objective rewards accurate transcription rather than the encoding of prosodic structure. Because prosodic boundaries and prominence are rarely annotated in such corpora, models receive little direct pressure to preserve these cues. This interpretation is consistent with recent LALM evaluations showing that failures can arise from perceptual limitations as well as downstream reasoning difficulties (Sakshi et al., 2025). Second, the audio encoders used by several evaluated models, such as Whisper-large-v2 in Qwen2-Audio and Whisper-large-v3-turbo in Ultravox, are optimized primarily for transcription and may discard prosodic detail before the representation reaches the language-model backbone. Third, modality-bridging layers may further compress prosodically distinct utterances into similar token-level representations, effectively reducing audio input to something closer to a transcript. Finally, strong syntactic priors inherited from the text-based backbone may dominate weak acoustic evidence. The persistence of a low-attachment preference even under prosodically HA-biased conditions is consistent with this possibility. Similar issues

have been reported by [Cheng et al. \(2025a\)](#), who showed that LALMs can rely on question content or prior expectations rather than the relevant acoustic cues.

In contrast to prosody, semantic cues exerted a stronger influence on model behavior, although the same asymmetry persists. Models reliably follow semantic bias when it favors low attachment but exhibit less consistent responses when semantic information favors high attachment. Rather than overriding the default low-attachment preference, semantic information appears to interact with it. This contrasts with human sentence processing, where comprehenders can revise an initial syntactic analysis when it leads to an implausible interpretation through semantic override ([Scheinberg et al., 2025](#); [Trueswell et al., 1994](#)). The present findings suggest that, although LALMs encode semantic knowledge, they do not flexibly integrate semantic plausibility with syntactic structure during ambiguity resolution. Instead, semantic information appears to modulate an established structural preference.

The comparison between prompt formats provides important methodological insight. Prior work ([Scheinberg et al., 2025](#)) showed that LLMs’ attachment preferences are sensitive to prompt formulation, with syntactic prompts encouraging structural heuristics and explicit A/B questions promoting greater reliance on semantic plausibility. Consistent with these findings, forced-choice prompting in this study reduces uncertainty compared with binary yes/no verification, suggesting that requiring an explicit attachment decision encourages models to reveal a preferred interpretation rather than leaving the ambiguity unresolved. Nevertheless, the persistence of the LA bias across both prompt formats indicates that this preference is not simply an artifact of response uncertainty, but reflects a stable characteristic of model behavior. Prompt design can therefore influence how models express their preferences without fundamentally changing the underlying attachment pattern.

This study also showed implications for modeling sentence processing using LALMs. Our findings suggest that LALMs do not exhibit human-like mechanisms for resolving syntactic ambiguity. These findings do not align with LALMs benchmarks that investigate content understanding such as (e.g., [Wang et al., 2025a](#); [Yang et al., 2024](#); [Ao et al., 2024](#); [Gao et al., 2025](#)) that report LALMs’ poor performance on non-English languages, roleplay scenarios, and phonetic ambiguity, but with benchmarks examining audio hallucinations (e.g., [Cheng et al., 2025a](#); [Ghosh et al., 2023](#)). We also claim that while models can exhibit surface-level alignment with human preferences in certain conditions (e.g., low-attachment bias), the underlying mechanisms do not necessarily reflect those of human comprehenders. This claim is consistent with findings showing that LLMs’ explicit metalinguistic judgments are dissociated from their implicit linguistic knowledge, suggesting that prompted responses should not be equated with models’ underlying linguistic generalizations ([Song et al., 2025](#)). In sum, the present study demonstrates that LALMs exhibit constrained and asymmetric cue integration in relative clause attachment. These results underscore both the potential and the limitations of LALMs in exhibiting human-like language processing capabilities.

6 Limitations and future directions

Several limitations should be noted. First, the language sample does not fully balance cross-linguistic attachment preferences. English and Chinese are described as low-attachment languages, while findings for Arabic are mixed. Thus, the present language set lacks a language with a robust high-attachment preference, leaving open whether the observed low-attachment bias partly reflects the sampled languages. Future work should include a balanced pool of low- and high-attachment languages, such as Spanish or French. Second, the prosodic stimuli were generated with TTS rather than natural human speech. Although acoustic verification confirmed the intended prosodic manipulations, the Chinese stimuli primarily realized prominence rather than silent boundary cues. Future work should validate the findings using naturally produced speech. Third, we compared LALM behavior with findings from the psycholinguistic literature rather than collecting parallel human judgments on the same stimuli. Direct human-model comparisons would provide a stronger benchmark. Finally, representation probing could help determine whether prosodic information is attenuated in the speech encoder, the audio-text projection layer, or the language-model backbone.

Reproducibility Statement

The dataset and code are publicly available at Hugging Face <https://huggingface.co/datasets/clap-purdue/MultiWhoAudio> and GitHub <https://github.com/clap-purdue/RC-LALMs>

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A Automatic Acoustic Verification of Prosodic Manipulations

We conducted an acoustic verification analysis to confirm that the intended prosodic manipulations (boundary placement and prominence) were realized in the generated audio. In the following, we report our methodology and results.

A.1 Sampling

For each of the three languages (English, Chinese, Arabic), we randomly sampled 15 sentence triplets, each consisting of a baseline (ambiguous) sentence and its HA-prosody and LA-prosody variants, using a fixed random seed (42) for reproducibility. This yielded 45 audio files per language and 135 files in total. The list of sampled item IDs is included in the project repository.

A.2 Measurement procedure

Audio files were analyzed using Parselmouth (Jadoul et al., 2018), a Python interface to the Praat acoustic analysis toolkit (Boersma et al., 2026). Word- and token-level boundaries were obtained via Whisper-based forced alignment using the stable-ts library (<https://github.com/jianfch/stable-ts>), with the large-v3 model (Radford et al., 2023). For each audio file, we extracted three classes of measures targeting the two intended prosodic manipulations (boundary placement and prominence).

First, **pause duration** at the intended boundary locations was measured using Praat’s silence detector, which identifies silent intervals relative to the utterance’s intensity maximum. The durations of silent intervals falling within each boundary region were summed. We measured the boundary after NP1 (relevant to the LA manipulation) and the boundary at the relative clause onset (relevant to the HA manipulation). For Chinese, whose relative clauses are prenominal, the RC-onset boundary was located at the relativizer; for Arabic, it was located at the relative pronoun. Second, **peak F0** on the head nouns of NP1 and NP2 was extracted and converted to semitones relative to each utterance’s median F0. Third, **mean intensity** on the head noun of each NP was computed in dB. To handle script-specific tokenization, target nouns were localized by exact token matching for English and by substring matching over a concatenated character stream for Chinese and Arabic. In addition, for Arabic, diacritics were stripped before matching.

A.3 Predicted patterns

Comparing the HA-prosody and LA-prosody conditions within matched triplets, we expect: (1) a longer boundary pause before the RC in HA; (2) a longer boundary pause after NP1 in LA; (3) higher peak F0 on NP1 in HA and on NP2 in LA; and (4) higher intensity on NP1 in HA and on NP2 in LA.

A.4 Statistical analysis

For each language and measure, we computed paired Wilcoxon signed-rank tests comparing HA and LA within matched triplets. We reported sample size n (the number of triplets for which the relevant span was successfully localized in both conditions), condition medians, the test statistic W , the p -value, and rank-biserial effect size r_{rb} .

A.5 Results

The verification confirms that the intended prosodic manipulations were realized in the generated stimuli. The manner of realization differs across languages in linguistically interpretable ways. Mean acoustic measures by language and condition are reported in Table 2, and paired Wilcoxon signed-rank tests comparing the HA and LA conditions within matched triplets are reported in Table 3.

Language	Condition	n	Pause before RC (ms)	Pause after NP1 (ms)	NP1 F0 (st)	NP2 F0 (st)	NP1 Intensity (dB)	NP2 Intensity (dB)
English	baseline	15	0	3	2.56	1.32	68.5	67.3
	HA	15	502	8	3.60	1.01	68.0	63.4
	LA	15	0	664	1.69	6.09	62.8	68.1
Chinese	baseline	15	30	0	10.23	6.35	63.6	66.1
	HA	15	21	0	6.20	9.35	66.3	61.1
	LA	15	45	0	13.64	6.22	66.5	67.4
Arabic	baseline	15	106	0	7.59	7.13	72.8	74.3
	HA	15	464	59	10.27	9.28	71.9	70.5
	LA	15	321	312	5.55	10.55	72.6	62.9

Table 2: Mean acoustic measures by language and condition. Pause durations are in ms, F0 in semitones relative to utterance median, and intensity is in dB.

Language	Comparison	Med. HA	Med. LA	W	p	r_{rb}	n
English	Pause before RC: HA > LA	470	0	0.0	< .001	1.00	15
	Pause after NP1: LA > HA	0	690	0.0	< .001	1.00	15
	NP1 peak F0 (st): HA > LA	3.07	0.99	44.0	.389	0.63	15
	NP2 peak F0 (st): LA > HA	0.99	6.65	0.0	< .001	1.00	13
	NP1 intensity (dB): HA > LA	68.32	64.35	0.0	< .001	1.00	15
	NP2 intensity (dB): LA > HA	63.87	67.68	5.0	< .001	0.96	15
Chinese	Pause before RC: HA > LA	21	45	—	—	—	15
	Pause after NP1: LA > HA	0	0	—	—	—	15
	NP1 peak F0 (st): HA > LA	2.30	7.48	37.0	.208	0.69	15
	NP2 peak F0 (st): LA > HA	1.26	5.57	20.0	.022	0.83	15
	NP1 intensity (dB): HA > LA	66.43	67.35	52.0	.679	0.57	15
	NP2 intensity (dB): LA > HA	62.89	66.11	14.0	.007	0.88	15
Arabic	Pause before RC: HA > LA	464	321	—	—	—	12
	Pause after NP1: LA > HA	0	240	2.0	.012	0.96	9
	NP1 peak F0 (st): HA > LA	8.67	5.55	7.0	.009	0.91	12
	NP2 peak F0 (st): LA > HA	8.34	9.25	30.0	.519	0.62	12
	NP1 intensity (dB): HA > LA	71.95	73.64	35.0	.791	0.55	12
	NP2 intensity (dB): LA > HA	70.46	62.43	7.0	.009	0.91	12

Table 3: Paired Wilcoxon signed-rank tests, HA vs. LA, by language.

Prominence cues were reliably realized in all three languages. In English, prominence on NP2 was strongly realized in the LA condition through both F0 (median 6.65 vs. 0.99 st, $p < .001$) and intensity ($p < .001$), and NP1 intensity prominence was strongly realized in HA ($p < .001$; Table 3). In Arabic, NP1 F0 prominence in HA ($p = .009$) and NP2 intensity

prominence in LA ($p = .009$) were both reliable. In Chinese, NP2 F0 prominence ($p = .022$) and NP2 intensity prominence ($p = .007$) were reliably realized in the LA condition. Where individual prominence comparisons did not reach significance (e.g., NP1 F0 in English, NP2 F0 in Arabic), effects were in the predicted direction with large effect sizes.

Boundary (pause) cues were realized in English and Arabic but not in Chinese. In English, both boundary manipulations were strongly realized. Considerable silence before the RC in HA (median 470 ms vs. 0 ms in LA) and after NP1 in LA (690 ms vs. 0 ms in HA), both $p < .001$ (Table 3). In Arabic, both pauses were realized in the predicted direction, with a clear pause after NP1 in LA (312 ms vs. 59 ms in HA) and a longer pre-RC pause in HA than in LA (464 vs. 321 ms; see Table 2). In Chinese, by contrast, pause durations were negligible across all conditions (≤ 45 ms; Table 2), well below the threshold of a perceptually meaningful prosodic boundary, and did not follow the predicted pattern. Because the same silence-detection procedure recovered correctly directed pauses in English and Arabic, we flag as a limitation for future investigation the possibility that the TTS system did not render the Chinese boundary manipulation as silent pausing. The Chinese manipulation was instead carried by prominence, consistent with the F0 and intensity results above. We note this as a stimulus-level limitation for Chinese in the study.

Overall, prominence cues were present in all three languages and boundary cues in two of three, with the cues most central to attachment disambiguation demonstrably realized. Crucially, the acoustic manipulations were robustly present in English, where the manipulation was most fully realized across both cue types, yet models there showed the same persistent LA bias reported in the study.

B Model Details

Table 4 provides a summary of the model details used in this study. All three models are open-source and support audio input for generating textual outputs. For further details on model architecture, training data, and evaluation, we refer the reader to the respective technical reports (Chu et al., 2024; Abouelenin et al., 2025) for Qwen2-Audio-Instruct and Phi-4-multimodal-instruct, respectively. For Ultravox v0.5, readers can check the GitHub repository at <https://github.com/fixie-ai/ultravox>.

Model	Details
Qwen2-Audio-Instruct	Qwen2-Audio-7B-Instruct is the instruction-tuned variant of Qwen2-Audio, built on the same Whisper-large-v2 encoder and Qwen-7B backbone architecture but further optimized for interactive dialogue through supervised fine-tuning on spoken dialogue data and instruction-following datasets.
Phi-4-multimodal-instruct	Phi-4-multimodal-instruct is a lightweight open-source multimodal foundation model developed by Microsoft, comprising 5.6 billion parameters and built upon the Phi-4-Mini-Instruct language model backbone with advanced encoders and adapters for vision and speech modalities. The model was trained end-to-end across all three modalities, meaning that text, vision, and audio inputs are processed by the same neural network rather than through a cascaded pipeline.
Ultravox v0.5	Ultravox v0.5 is a multimodal speech language model that integrates a pretrained Llama-3.1-8B-Instruct backbone with the encoder component of Whisper-large-v3-turbo to process both speech and text inputs and generate textual outputs.

Table 4: Model details.

C Model Response Examples

Rather than using semantic information to resolve ambiguity, models sometimes treat such cases as underspecified, either requesting additional context or declining to commit to a specific interpretation. In some cases, this behavior reflects alignment constraints, as models may refuse to answer questions they interpret as ethically problematic, indicating that safety considerations can override interpretive decisions. Moreover models tend to require explicit information to justify an interpretation, rather than inferring a plausible one. When semantic cues conflict with structural preferences, models often either remain uncertain or revert to default attachment strategies. Their explanations frequently appeal to structural heuristics such as locality-based attachment (i.e., linking the relative clause to the nearest noun), consistent late closure or recency principle (i.e., the human parser prefers to attach new incoming information to the phrase currently being processed). This suggests that attachment decisions are driven by learned syntactic regularities, potentially reflecting training data dominated by English. Table 5 illustrates representative examples of these behaviors. Overall, although LALMs are sensitive to semantic cues, they do not integrate them in a human-like manner; instead, they rely on rigid structural heuristics or treat ambiguity as insufficient information, revealing a failure of semantic override.

Behavior Type	Model Response Example
Treating ambiguity as underspecified or requesting additional context	The specific person chosen as king is not mentioned in the provided information, so the answer cannot be determined with certainty. It is possible that the wife or a relative could have been chosen, but without further context or information, it is impossible to determine who was ultimately selected as king.
Alignment constraints or refusing ethically problematic interpretations	I’m sorry, but I cannot answer this question as it is based on a stereotype and does not respect the privacy of individuals. It is important to treat all individuals with respect and dignity, regardless of their menstrual status.
Reverting to default structural heuristics	Based on the sentence structure, “Robert yelled at the assistant of the dentist who had examined thousands of teeth,” the relative clause “who had examined thousands of teeth” most naturally modifies the nearest preceding noun, which is “the dentist.” In English grammar, this is known as “low attachment.”

Table 5: Examples of model responses to ambiguous sentences.